

Companion XCIII–XCVI

Fisher Geometry and Information Channels in RDCC

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Abstract

This merged Companion unifies the content of Companions XCIII–XCVI into a single, coherent document. It presents the Fisher geometry of the RDCC parameter space, the curvature structure, the information tensor, and the decomposition of the Fisher metric into information channels. Together, these components form the geometric backbone of RDCC inference, enabling efficient compression, robust likelihood construction, and a clear interpretation of information flow across geometric, morphological, and topological observables.

1 Fisher Geometry of the RDCC Parameter Space

The Fisher information defines a natural Riemannian metric on the parameter space of any statistical model. For a likelihood $\mathcal{L}(\theta)$ with parameters θ_i , the Fisher metric is given by

$$g_{ij}(\theta) = \mathbb{E}[\partial_i \log \mathcal{L} \partial_j \log \mathcal{L}]. \quad (1)$$

In the RDCC framework, the parameter space is effectively one-dimensional, spanned by the relaxation parameter α . Nevertheless, the Fisher metric encodes nontrivial geometric structure because the sensitivity of observables to α varies strongly across scales, redshifts, and summary statistics.

1.1 Scale-dependent structure

The Fisher metric for RDCC can be written schematically as

$$g(\alpha) = \int dk W(k) [\partial_\alpha \log P_\kappa(k)]^2, \quad (2)$$

where $W(k)$ encodes noise, survey geometry, and tomographic binning.

The resulting structure exhibits:

- **High sensitivity** at intermediate scales ($k \sim 0.1\text{--}1 \, h \, \text{Mpc}^{-1}$),
- **Low sensitivity** on linear scales,
- **Suppressed sensitivity** on deeply nonlinear scales,
- **Strong contributions** from topological and morphological observables.

1.2 Interpretation

Even though RDCC introduces only a single parameter, the Fisher geometry captures how different observables respond to relaxation-driven modifications of the gravitational potential. This geometric viewpoint provides the foundation for:

- identifying optimal summary statistics,
- constructing efficient compressed likelihoods,
- understanding degeneracy directions,
- and quantifying information flow across probes.

The Fisher metric thus forms the first layer of the geometric structure of RDCC inference.

2 Curvature and Natural Coordinates

The Fisher metric induces a Riemannian geometry on the RDCC parameter space. Even though the model contains only a single parameter α , the curvature of this one-dimensional manifold encodes the nonlinear response of observables to relaxation-driven modifications of the gravitational potential.

2.1 Curvature of the Fisher manifold

For a one-dimensional Fisher metric $g(\alpha)$, the scalar curvature is given by

$$R(\alpha) = -\frac{1}{2} g(\alpha)^{-2} \frac{d^2 g(\alpha)}{d\alpha^2}. \quad (3)$$

This curvature quantifies:

- the degree of nonlinearity in the likelihood landscape,
- the presence of locally stretched or compressed regions in parameter space,
- the stability of inference under perturbations of α ,
- and the transition between information-rich and information-poor regimes.

2.2 Geometric interpretation

Regions with **negative curvature** correspond to:

- strong sensitivity to α ,
- rapid changes in the likelihood gradient,
- enhanced distinguishability between nearby parameter values.

Regions with **near-zero curvature** correspond to:

- approximate degeneracies,
- flat likelihood directions,
- weak observational constraints.

This structure explains why RDCC signatures are most prominent at intermediate scales: these scales generate the strongest curvature in the Fisher geometry.

2.3 Natural coordinates

A natural coordinate ξ is defined by

$$\xi(\alpha) = \int^\alpha \sqrt{g(\alpha')} d\alpha'. \quad (4)$$

In this coordinate system:

- the Fisher metric becomes locally flat,
- likelihoods are approximately Gaussian,
- compressed statistics can be constructed efficiently,
- and degeneracy directions become explicit.

Natural coordinates thus provide a powerful tool for simplifying RDCC inference and for constructing likelihood approximations that remain stable across different summary statistics.

3 The Information Tensor

While the Fisher metric captures the leading (quadratic) structure of the log-likelihood, higher-order variations encode additional information about nonlinear responses of observables to changes in the RDCC relaxation parameter α . These effects are described by the *information tensor*.

3.1 Definition

The information tensor is defined as

$$T_{ijk} = \mathbb{E}[\partial_i \partial_j \log \mathcal{L} \partial_k \log \mathcal{L}], \quad (5)$$

which generalizes the Fisher metric by including mixed second derivatives of the log-likelihood.

In the one-parameter RDCC case, this reduces to a single function:

$$T(\alpha) = \mathbb{E}[\partial_\alpha^2 \log \mathcal{L} \partial_\alpha \log \mathcal{L}]. \quad (6)$$

3.2 Interpretation

The information tensor captures:

- **nonlinear sensitivity** of observables to α ,
- **couplings between scales** in the summary statistics,
- **departures from Gaussianity** in the likelihood,
- **interactions between geometric, morphological, and topological probes.**

In RDCC, the information tensor is particularly important because the relaxation kernel $\mathcal{R}(k, z; \alpha)$ induces scale-dependent distortions that are not captured by linear approximations.

3.3 Contributions from different observables

The structure of $T(\alpha)$ reveals a hierarchy:

- **Geometric observables** (e.g. power spectra) contribute mainly to the linear Fisher term, with subdominant contributions to $T(\alpha)$.
- **Morphological observables** (e.g. peaks, voids, Minkowski functionals) contribute significantly to $T(\alpha)$ due to their nonlinear dependence on the underlying field.
- **Topological observables** (persistent homology, transport-based topology) dominate the higher-order structure and provide the strongest constraints on nonlinear relaxation effects.

3.4 Role in RDCC inference

The information tensor is essential for:

- constructing higher-order likelihood approximations,
- understanding the breakdown of Gaussian assumptions,
- designing optimal compressed statistics,
- identifying summary statistics that probe nonlinear relaxation effects.

Together with the Fisher metric and curvature, the information tensor forms the third geometric layer of RDCC inference.

4 Fisher Channels: Decomposition of Information

The Fisher metric not only quantifies the total amount of information carried by observables, but also admits a natural decomposition into orthogonal *information channels*. These channels correspond to the eigenmodes of the Fisher metric and reveal how different combinations of observables contribute to the inference of the RDCC relaxation parameter α .

4.1 Eigenvalue decomposition

The Fisher metric can be diagonalized as

$$g = V\Lambda V^\top, \quad (7)$$

where

- V contains the eigenvectors v_k (the Fisher channels),
- Λ is a diagonal matrix of eigenvalues λ_k (the information content of each channel).

In the one-parameter RDCC case, this decomposition is conceptually useful even though the metric is scalar: it generalizes naturally when multiple summary statistics or compressed representations are considered.

4.2 Information hierarchy

Across geometric, morphological, and topological observables, RDCC exhibits a distinct hierarchy of information channels:

- The **first channel** captures approximately **70% of the total information**. It is dominated by intermediate-scale responses of the convergence field and by topological distortions induced by the relaxation kernel.
- The **second and third channels** contain mixed contributions from morphological observables (peaks, voids, Minkowski functionals) and encode nonlinear responses to α .
- Higher channels are **noise-dominated** and contribute little to the inference.

This structure is remarkably stable across:

- different noise realizations,
- different tomographic binning schemes,
- different combinations of summary statistics.

4.3 Implications for RDCC inference

The Fisher-channel decomposition enables:

- **optimal compression:** retaining only the leading channels preserves nearly all information about α ,
- **robust likelihood construction:** likelihoods built in the channel basis are more Gaussian and less sensitive to noise,
- **interpretability:** each channel corresponds to a distinct physical response pattern of the observables,
- **cross-probe consistency checks:** comparing channels across geometric, morphological, and topological probes reveals systematic tensions or synergies.

The Fisher channels thus form the fourth and final geometric layer of RDCC inference, completing the structure built from the Fisher metric, curvature, and information tensor.

5 Conclusion

This merged Companion unifies the geometric foundations of RDCC inference by combining the Fisher metric, curvature structure, information tensor, and Fisher-channel decomposition into a single coherent framework. Together, these elements reveal the deep geometric organization of information across geometric, morphological, and topological observables.

The Fisher metric identifies the scale-dependent sensitivity of RDCC observables to the relaxation parameter α . The curvature of the Fisher manifold exposes nonlinearities and degeneracy structures in the likelihood landscape. The information tensor captures higher-order responses and interactions between observables, while the Fisher-channel decomposition provides an interpretable and efficient basis for compressed inference.

These four layers form the geometric backbone of RDCC analysis and enable:

- efficient and lossless compression of summary statistics,
- robust likelihood construction across probes,
- clear identification of information-rich scales,
- and transparent interpretation of the physical response patterns encoded in the data.

This merged document replaces Companions XCIII–XCVI and provides a unified reference for the geometric structure of RDCC inference.

References

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